

gitops-reverser in CozyStack?

COZYSTACK



Reverse GitOps for platform changes

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Who am I?

[Software|Product|Cloud|AI] engineer

- Worked in startups, consultancy, and SaaS
- Left my job last summer to pursue building my own company



What do I do?

- koudijs.dev
 - consultancy, training
- [ConfigButler](#)
 - startup, open source first
 - helps you build high quality configuration
- [Reverse GitOps](#) pioneer
 - manifest, feel free to comment! 😊

Tenant /

root



Deployed applications

Tenant root · tenant-root

NATS 4

**lastone**

NATS

● Ready

1h

**my-first-again**

NATS

● Ready

5d

**test123**

NATS

● Ready

5d

**test1234**

NATS

● Ready

1d

Why these resources?

- 🕒 When? ✅
- 👤 Who? ❌
- 🤔 Why? ❌

Git seems far away.

Your options

- Accept it
- Periodic reset
- Limited access
- Audit files ([kube-api](#) to the rescue)

Your options: GitOps

Flip the interface: **Git is the only way in.**

- YAML files / PRs
- Controllers (Argo, Flux)

But: edits can't go through kubectl, dashboard, MCP. The GUI and the API stop being the front door. Git is your **interface**.

PAAS

- Click House
- Foundation DB
- Harbor
- Kafka
- Maria DB
- Mongo DB
- NATS
- Open BAO
- Postgres
- Qdrant
- Rabbit MQ
- Redis

BACKUPS

- Plans
- Backup Jobs
- Backups
- Restore Jobs

ADMINISTRATION

- Info
- Modules
- External IPs

Tenant / root

Deployed applications

Tenant root · tenant-root

NATS 4

 **lastone**
NATS Ready 1h

 **my-first-again**
NATS Ready 5d

 **test123**
NATS Ready 5d

 **test1234**
NATS Ready 1d

[+ Deploy application](#)

Where we are now

We have seen three things:

1. Kubernetes audit events can show **exactly what happened**
2. gitops-reverser can turn those events into **readable Git history**
3. The coffee-app demo made that concrete in a Cozystack-shaped workflow

Commits

🔗 main	👤 All users
🕒 Commits on May 26, 2026	
[CREATE] examples.configbutler.ai/v1alpha1/quizsubmissions/cozysummit-warmup-8dr67 👤 Anonymous 186362 authored and configbutler-bot committed 2 hours ago	Verified
[CREATE] examples.configbutler.ai/v1alpha1/quizsubmissions/cozysummit-feedback-frbf4 👤 Anonymous 146001 authored and configbutler-bot committed 2 hours ago	Verified
[CREATE] examples.configbutler.ai/v1alpha1/quizsubmissions/cozysummit-warmup-dgm2x 👤 Anonymous 146001 authored and configbutler-bot committed 2 hours ago	Verified
reconcile: sync 6 resources 👤 configbutler-bot committed 3 hours ago	Verified
[UPDATE] examples.configbutler.ai/v1alpha1/coffeeconfigs/app-config (#1) 👤 sunib authored 3 hours ago	Verified
[UPDATE] examples.configbutler.ai/v1alpha1/coffeeconfigs/app-config 👤 sunib authored and configbutler-bot committed 3 hours ago	Partially verified
Normal shopname	10

First proof: audit capture

The audit log gives the missing context:

- actor from OIDC
- verb and resource
- request and response
- timestamp from the API server

That is enough to write an attributed commit.



configbutler-bot

ConfigButler bot

Committed to this repository in the past month

Follow

 **unib** authored and **configbutler-bot** committed on May 22 Partially verified

[UPDATE] apps.cozystack.io/v1alpha1/natses/andmore3

 main

1 parent 146ddd commit c641430

1 file changed +20

Filter files...

demo/apps.cozystack.io/v1...

andmore3.yaml

Search within code

demo/apps.cozystack.io/v1alpha1/natses/tenant-root/andmore3.yaml +20

@@ -0,0 +1,20 @@

1 + apiVersion: apps.cozystack.io/v1alpha1

2 + kind: NATS

3 + metadata:

4 + annotations:

5 + simon: was here, and again

6 + name: andmore3

7 + namespace: tenant-root

8 + spec:

9 + config:

10 + merge: {}

11 + resolver: {}

12 + external: false

13 + jetstream:

14 + enabled: true

15 + size: 16i

16 + replicas: 2

17 + resources: {}

18 + resourcesPreset: t1.nano

19 + storageClass: local

20 + users: {}

Comments 0

Lock conversation

12

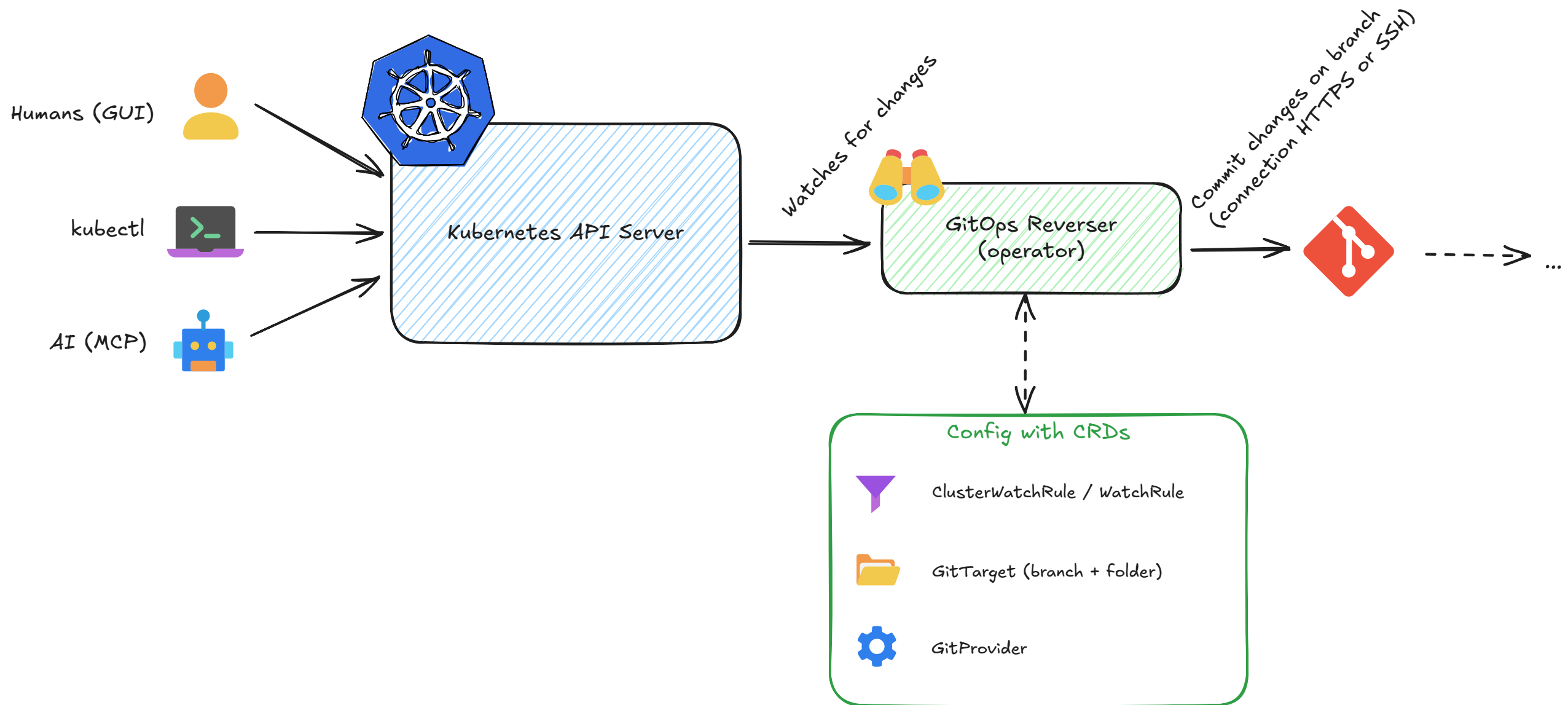
Reverse GitOps

Classic GitOps: **Git -> Platform**

Reverse GitOps: **Platform -> Git**

Git stops being the interface.

Git becomes the memory.



Then: the coffee app

Together we used the coffee-app demo:

- create higher-level platform resources
- change configuration through the normal interface
- resolve merge conflicts as we go

STOREFRONT CONFIG

GENERATION 13

SHOP NAME

CozySummit Coffee1



CURRENCY

EUR

BANNER TEXT

Scan the QR code for your free CozySummit coffee

PRODUCTS

Add Product

SKU

coffee-flat-white

NAME

Flat White

BASE PRICE (CENTS)

395

☒ Enabled

DESCRIPTION

Double shot with silky milk

€3.95

Remove

SIGNED
IN

REVIEW AND SAVE

1 CHANGE(S) READY TO SAVE

Yellow dots are local edits. Saving writes your changes into the watched config.

shopName

EDITED

Revert

~~CozySummit Coffee~~ → CozySummit Coffee1

WHY ARE YOU MAKING THIS CHANGE?

Optional for now. Add a short note so people understand what changed and why.

1 edit(s) and 0 missed incoming change(s) in this save.

Save 1 Change

What we learned

- We can answer **when**
- We can answer **who**
- With good resource diffs, we can often answer **why**

And we can do that without making Git the only front door.

AC

admin@reverser-cozy

▼

apps.foundationdb.org

▼

autoscaling.k8s.io

▼

backups.cozystack.io

▼

cert-manager.io

▼

cilium.io

▼

clickhouse-keeper.altinity.com

▼

clickhouse.altinity.com

▼

configbutler.ai

▲

Cluster Watch Rule

Commit Request

Git Provider

Git Target

Watch Rule

core.strimzi.io

▼

GitTarget

3 items

All namespaces

GitTarget search ...

☐	Name	Namespace	Provider	Branch	Path	Ready	Status	Age
☐	cozystack-example	gitops-reverser	cozystack-example	main	demo	True	All lifecycle gates sati...	6d2h
☐	voter-demo	voter-production	voter-demo	main	voter	True	All lifecycle gates sati...	25h
☐	voter-demo	voter-test	voter-demo	features/some-feature	voter	True	All lifecycle gates sati...	25h

New feature idea

For every captured resource, serve provenance back by API:

```
GET /resources/{group}/{version}/{resource}/{namespace}/{name}
```

```
{  
  "gitPath": "clusters/demo/apps/coffee/order.yaml",  
  "lastCommitUrl": "https://git/.../commit/abc123",  
  "historyUrl": "https://git/.../commits/main/.../order.yaml"  
}
```

Why API, not annotations?

Annotations are tempting, but they make the reverser a writer in the cluster.

- every annotation write creates another event
- the operator has to filter its own changes
- live objects become coupled to Git layout

A read API keeps the mirror read-only.

Trust boundary

I want this to be a **trustable source**.

If a commit says "Noa changed the coffee config", that should mean:

- the API server authenticated Noa
- the audit event really came from that API server
- the commit author was derived from that trusted event

The annoying part: mTLS

The kube-apiserver audit webhook should use mutual TLS:

- apiserver verifies the reverser endpoint
- reverser verifies the apiserver client certificate
- only verified audit facts attach a human name to a commit

This is hard and annoying for easy rollout.

```
apiServer:  
  extraArgs:  
    audit-policy-file: /var/audit-policy.yaml  
    audit-webhook-config-file: /var/audit-webhook.yaml  
    audit-webhook-batch-max-wait: 1s  
    audit-webhook-batch-max-size: "100"
```

Is it worth it?

For generic Kubernetes: maybe not for everyone.

For Cozystack: I think yes.

- Cozystack owns the kube-apiserver
- the dashboard already uses the Kubernetes API
- high-level resources create human-readable diffs
- the distro can hide the difficult default setup

What I would propose

Start with a focused integration:

- read-only mirror of selected Cozystack resources
- OIDC-attributed commits from verified audit events
- default GitProvider and GitTarget
- GUI links to file location, latest commit, and history

No Git -> cluster write-back yet.

Open questions

- Which Cozystack resources should be captured first?
- Can `apps.cozystack.io/v1alpha1` secret fields be removed?
- Where should the default Git repository live?
- What should be visible per tenant?
- How much of the audit/mTLS setup can Cozystack ship as defaults?

Takeaways

1. Keep Cozystack API-first
2. Git as trustworthy memory
3. Make provenance visible from the GUI by adding a link
4. Treat mTLS as the price of trustworthy attribution

Contact details and presentation at <https://koudijs.dev>